

ASME
IDETC-CIE 2026

INTERNATIONAL DESIGN ENGINEERING TECHNICAL CONFERENCES & COMPUTERS AND INFORMATION IN ENGINEERING CONFERENCE

HILTON AMERICAS HOUSTON | HOUSTON, TX, USA

Team human-in-the-loop

ASME HACKATHON 2026

- Problem 3 (Autodesk)
- Agent Harness for CAD-Editing
- Technical University of Darmstadt, Germany



Problem

Limitations of Baseline Harness:

- Initial Starting position (only one view)
- No Version Control (zero shot script gerneration)
- Geometric grounding is missing

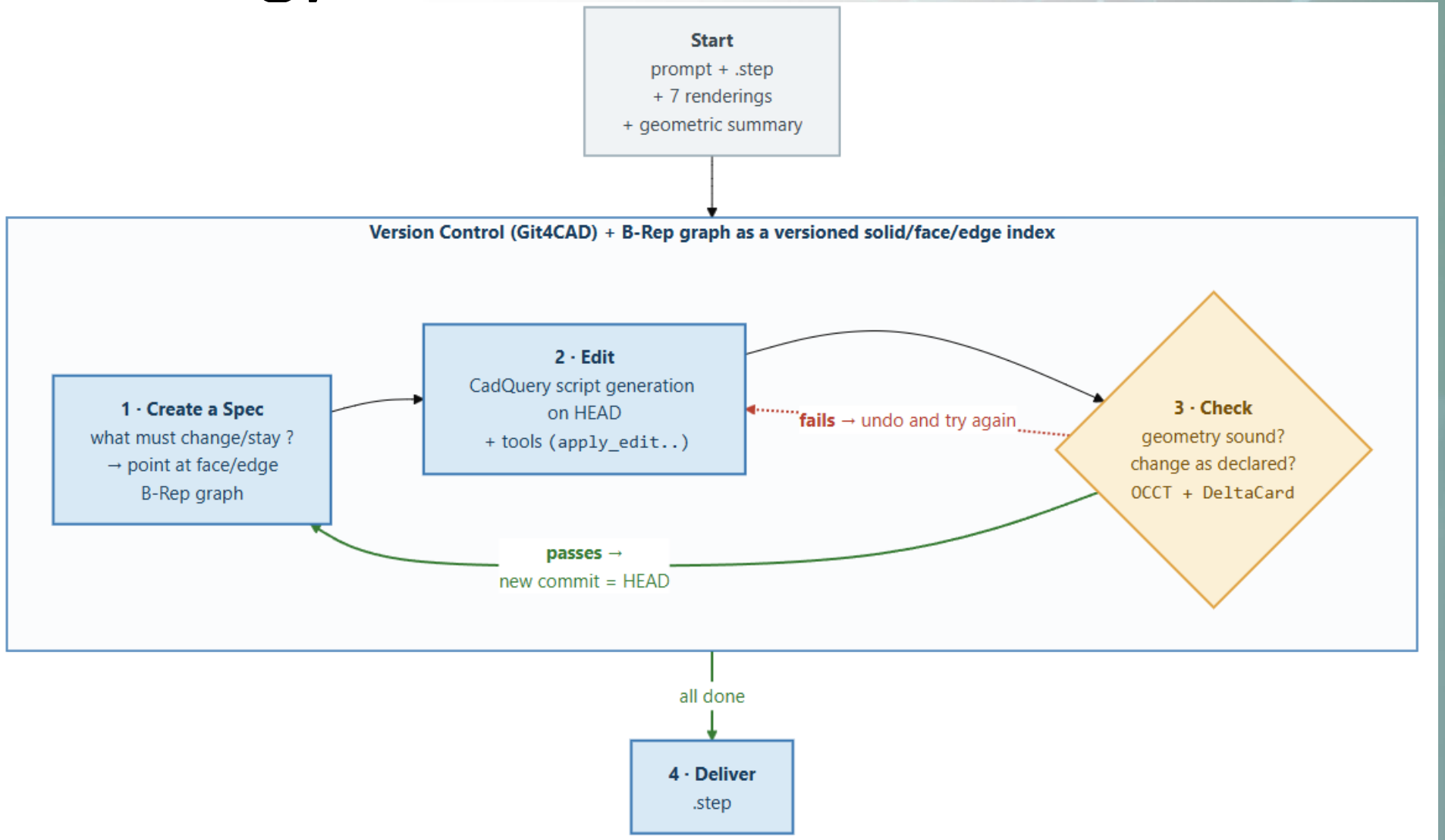


Methodology

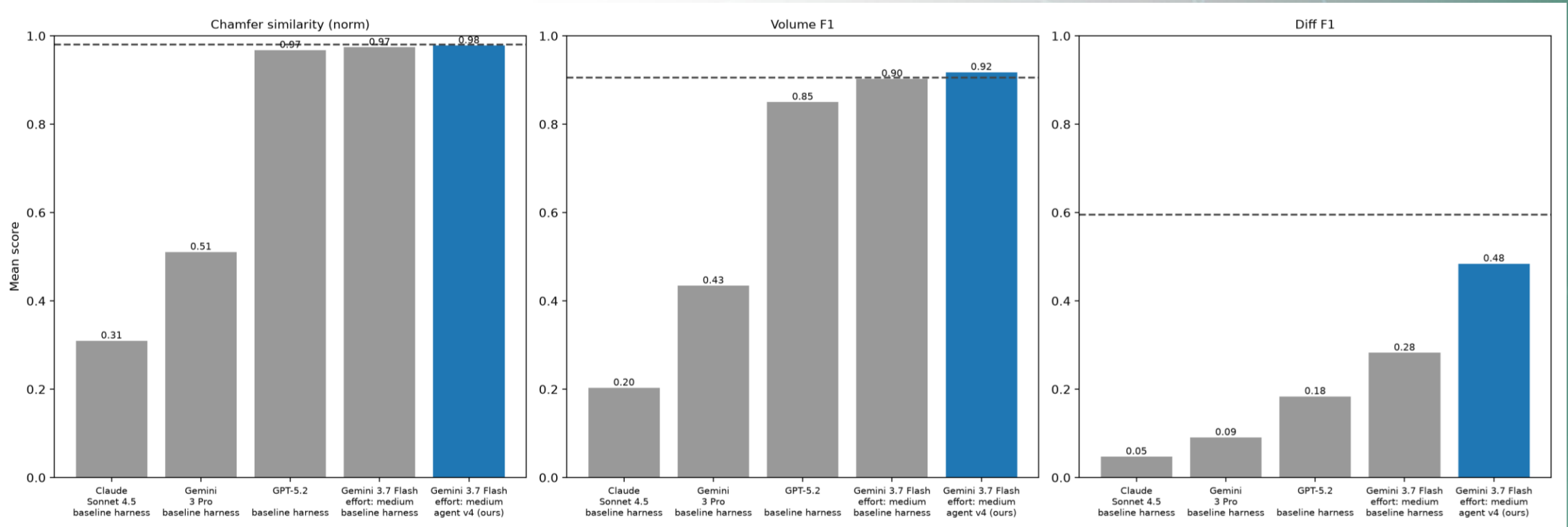
- Implement Version Control („Git4CAD“)
- Implement BREP-Graph for geometric grounding
- Implement Spec-driven design changes

- Compare: Baseline Harness vs Improved Harness (v4) under same Model (Gemini 3.7-flash) Effort Level (medium) and max.Iterations (10)

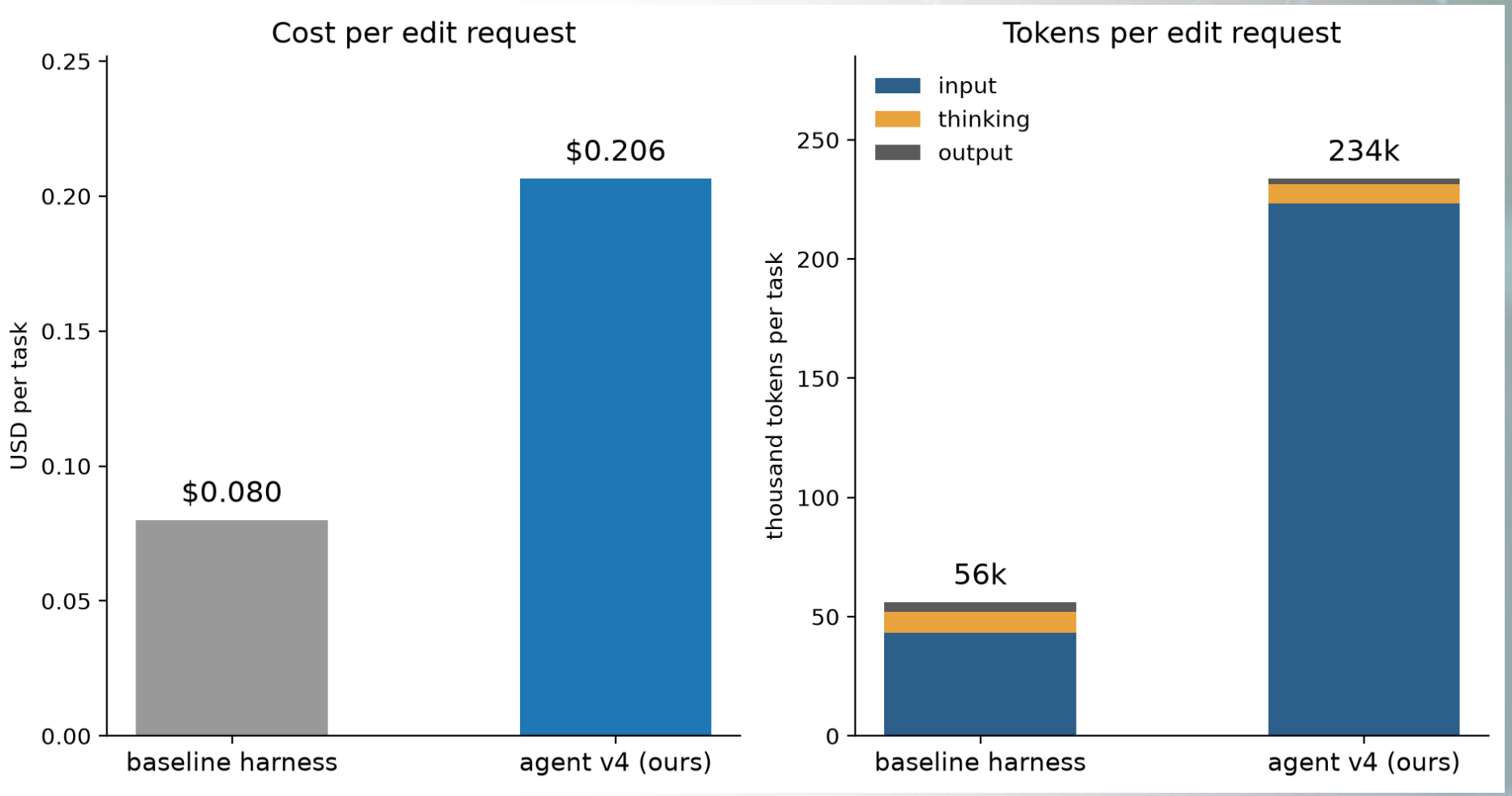
Methodology



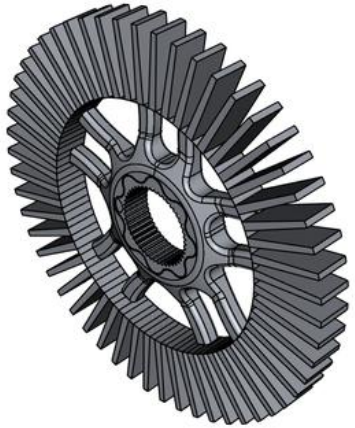
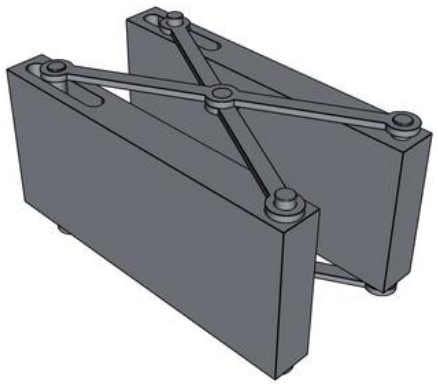
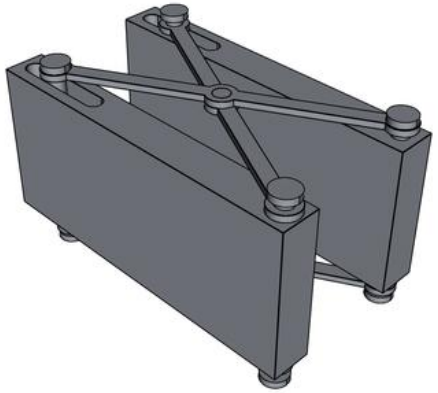
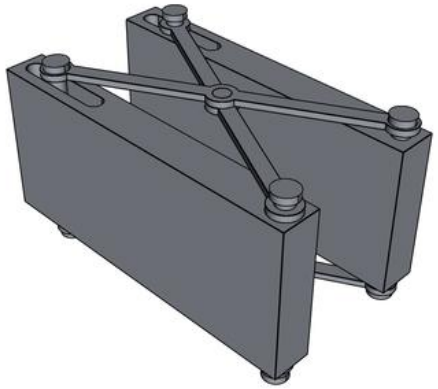
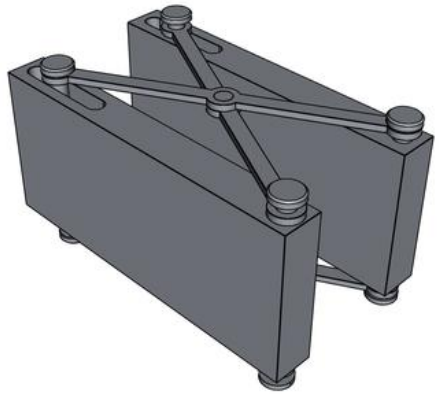
Results: Scores, automated metrics



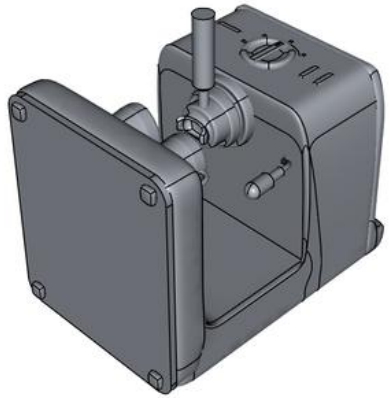
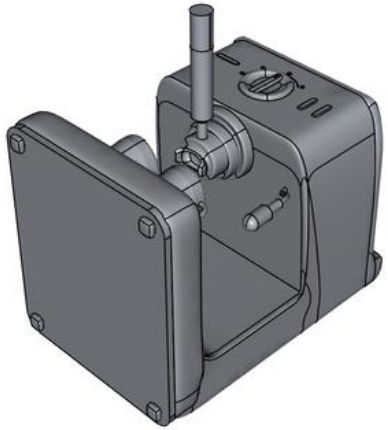
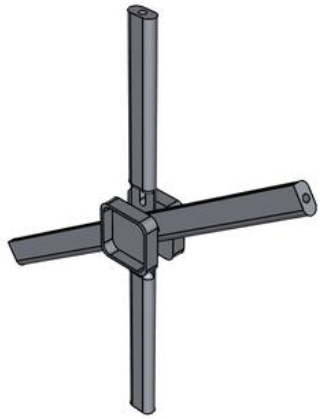
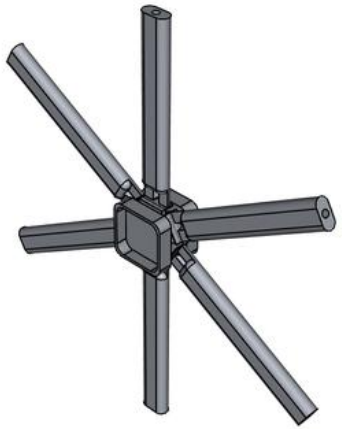
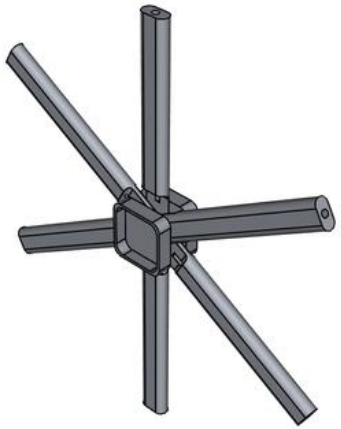
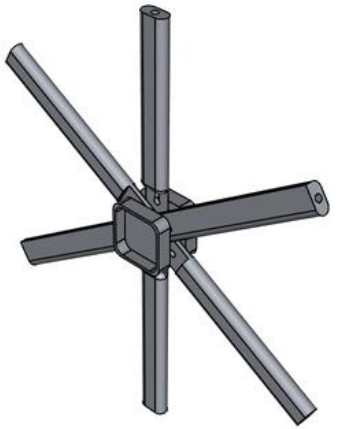
Results: Cost



Team human-in-the-loop (HITL)

	Start	Baseline harness Gemini 3.7 Flash, effort: medium	Agent v4 (ours) Gemini 3.7 Flash, effort: medium	Human ground truth
Medium Hi , Please convert the round edges of the gear into straight spur gear teeth.		 diff F1 0.019	 diff F1 0.000	
Medium Add cylindrical heads on the long pins to prevent link arms against slipping off.		 diff F1 0.399	 diff F1 0.756	

Team human-in-the-loop (HITL)

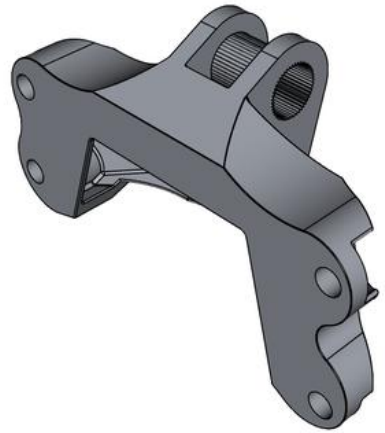
	Start	Baseline harness Gemini 3.7 Flash, effort: medium	Agent v4 (ours) Gemini 3.7 Flash, effort: medium	Human ground truth
Easy Prolong black lever sticking out to the front by 5cm for better manipulation.		 diff F1 0.710	 diff F1 0.798	
Hard Add third rotor blade to the assembly. It should have the same design as the other two. Radii should applied on all four long edges of the blade. Adjust central portion of all three blades so that there is a space for a new blade - reduce thickness of the central portion to cca 0,42mm. New blade should go through the center of the stack.		 diff F1 0.761	 diff F1 0.796	

Team human-in-the-loop (HITL)

ASME IDETC-CIE 2026
INTERNATIONAL DESIGN ENGINEERING TECHNICAL CONFERENCES & COMPUTERS AND INFORMATION IN ENGINEERING CONFERENCE

Hard

Add a connecting hole of 1.7 millimetre diameter and apply 0.1 millimetre grooves to increase grip and prevent slipping.

Start	Baseline harness Gemini 3.7 Flash, effort: medium	Agent v4 (ours) Gemini 3.7 Flash, effort: medium	Human ground truth
			
	diff F1 0.002	diff F1 0.073	

Outlook & Future Implementations

- Multi-Agent-Systems (Plan, Edit, Judge) if benefits > complexity
- Iterative Prompting (Plan mode or clearification questions) after vague prompts
- Memory to learn from edits and Feedback
- Multiple LLM (Ensemble Planing or Judgement via panel discussion and majority voting)
- Include generative Models such as AutoBrep